FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

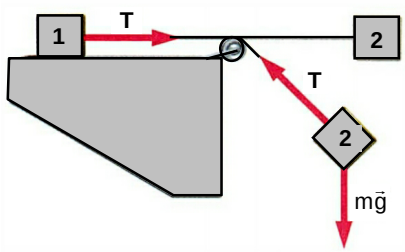
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The range for a projectile that lands at the same elevation from which it is fired is given by $R = (u^2/g) \sin 2\theta$. Assume that the angle of projection = 30°. If the initial speed of projection is increased by 1%, while the angle of projection is decreased by 2% then the range changes by
(A) -0.3% (B)+ 4.3%
(C) +0.65% (D) 0.85%

2. Two highways are perpendicular to each other: imagine them to be along the x-axis and the y-axis, respectively. At the instant t = 0, a police car P is at a distance d = 400 m from the intersection and moving at speed of 80 km/h towards it along the x-axis. Motorist M is at a distance of 600 m from the intersection and moving towards it at a speed of 60 km/h along the y-axis. The minimum distance between the cars is
(A) 300 m (B) 240 m.
(C) 180m. (D) 120 m

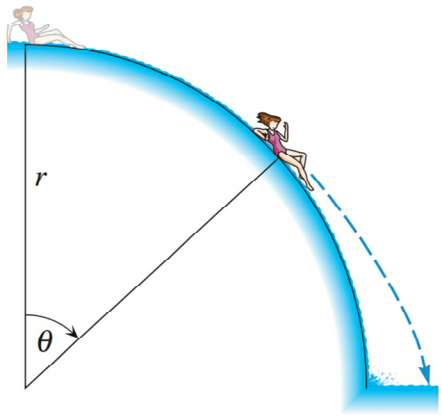
3. Two identical blocks are attached by a massless string running over a pulley as shown in Figure. The rope initially runs over the pulley at its (the rope's) midpoint, and the surface that block 1 rests on is frictionless. Blocks 1 and 2 are initially at rest when block 2 is released with the string taut and horizontal. Assume that the initial distance from block 1 to the pulley is the same as the initial distance from block 2 to the wall.



(A) Block 1 hits the pulley before Block 2 hits the wall
(B) Block 2 hits the wall before block 1 hits the pulley
(C) Block 1 hits pulley simultaneously with block 2 hitting the wall
(D) Which block hits the obstruction first depends on the actual mass of the blocks and the length of the string.

Space for Rough work

4. A person starts with a speed of $\sqrt{(0.5gr)}$ at the top of a large frictionless spherical surface, and slides into the water below (see the drawing). Then, the person
(A) loses contact when $\cos\theta = 1/3$
(B) slides through a height of r/6 before losing contact
(C) slides through a height of r/3 before losing contact
(D) never loses contact with the surface



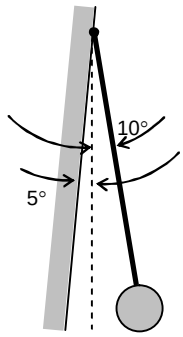
5. Suppose that water drops are released from a point at the edge of a roof with a constant time interval Δt between one water drop and the next. The drops fall a distance h to the ground. If Δt is very short ie the number of drops falling though the air at any given instant is very large then the CM of the drops is very nearly at a height (above the ground) of
(A) h/2 (B) h/3
(C) 2h/3 (D) 3h/4

6. The moment of inertia of a uniform solid regular tetrahedron of mass m and edge a, about its symmetry axis (i.e. an axis passing through one vertex and the centre of the opposite face) is:
(A) $ma^2/10$ (B) $3ma^2/10$
(C) $ma^2/15$ (D) $ma^2/20$

7. A uniform rod lies at rest on a frictionless horizontal surface. A particle, having a mass equal to that of the rod, moves on the surface perpendicular to the rod collides with it, and finally sticks to it. The minimum loss of KE in the collision, under the given conditions, is
(A) 5% (B) 10%
(C) 20% (D) 30%

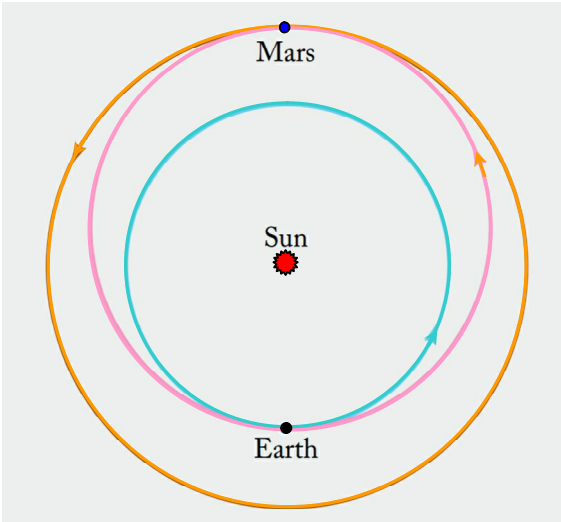
Space for Rough work

8. A pendulum of length 1m hangs from an inclined wall. Suppose that this pendulum is released at an initial angle of 10° and it bounces off the wall elastically when it reaches an angle of -5° as shown in the figure. Take $g = \pi^2$ m/s². The period of this pendulum is (in second)



(A) 2/3
(B) 3/2.
(C) 3/4.
(D) 4/3

9. The Earth has a circular orbit of radius r and period t around the Sun; Mars has a circular orbit of radius R and period T. In order to send a spacecraft from the Earth to Mars, it is convenient to launch the spacecraft into an elliptical orbit whose perihelion coincides with the orbit of the Earth and whose aphelion coincides with the orbit of Mars; this orbit requires the least amount of energy for a trip to Mars. The time t' taken by a spacecraft to reach Mars from the Earth satisfies:



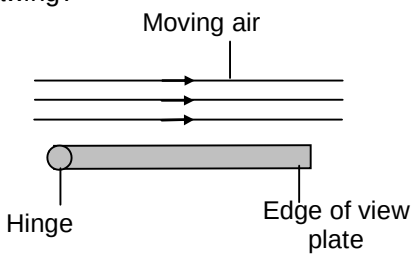
(A) $t'=(t + T)/2$
(B) $t'^2 = (t^2 + T^2)/2$
(C) $t'^{2/3} = (t^{2/3} + T^{2/3})/2$
(D) $(2t')^{2/3} = (t^{2/3} + T^{2/3})/2$

10. When a force accelerates a body immersed in a fluid, some of the fluid must also be accelerated, since it must be pushed out of the way of the body and flow around it. Thus, the force must overcome not only the inertia of the body, but also the inertia of the fluid pushed out of the way. It can be shown that for a spherical body completely immersed in a nonviscous fluid, the extra inertia is that of a mass of fluid half as large as the fluid displaced by the body. The acceleration of a small spherical air bubble in water is nearly

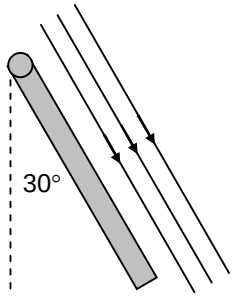
(A) zero.
(B) g
(C) g/2.
(D) 2g

Space for Rough work

11. A uniform rectangular plate is hanging vertically downward from a hinge that passes along its left edge. By blowing air at 12 m/s over the top of the plate only, it is possible to keep the plate in a horizontal position, as illustrated in part (a) of the drawing. To what value of speed should the air be blown so that the plate is kept at a 30° angle with respect to the vertical, as in part (b) of the drawing?



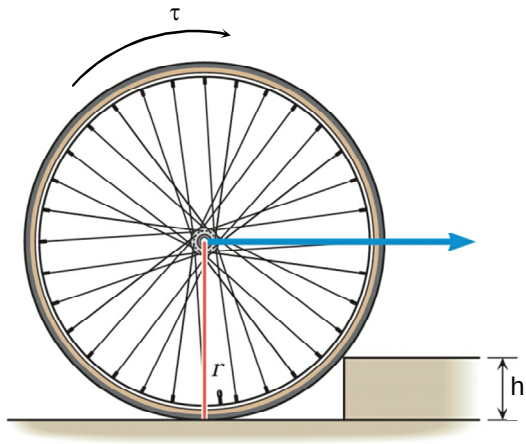
(a)



(b)

(A) 12m/s
(B) 6 m/s
(C) $6\sqrt{2}$ m/s
(D) $12\sqrt{2}$ m/s

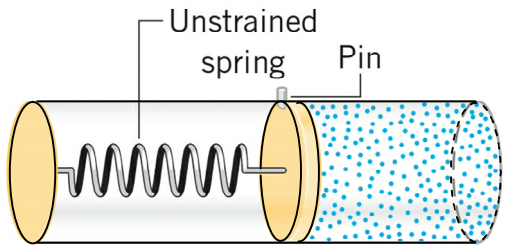
12. The drawing shows a bicycle wheel of radius r, resting against a small step whose height is h=r/5. A clockwise torque is applied to the axle of the wheel. As the magnitude of torque increases, there comes a time when the wheel just begins to rise up and loses contact with the ground. Let this torque be τ. What is the magnitude of the horizontal component of the acceleration of the centre of the wheel when a torque of 2τ is applied? Assume that the wheel doesn't slip at the edge of the step when this torque is applied (ignore the mass of the spokes).



(A) 4.8 m/s²
(B) 3 m/s²
(C) 2.4 m/s²
(D) 1.5 m/s²

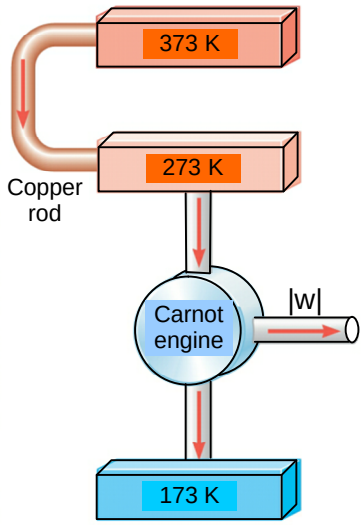
Space for Rough work

13. A gas fills the right portion of a horizontal cylinder whose radius is 5.00 cm. The initial pressure of the gas is 101 kPa. A frictionless movable piston separates the gas from the left portion of the cylinder, which is evacuated and contains an ideal spring, as the drawing shows. The piston is initially held in place by a pin. The spring is initially unstrained, and the length of the gas-filled portion is 20.0 cm. When the pin is removed and the gas is allowed to expand, the length of the gas-filled chamber doubles when it finally reaches equilibrium. The initial and final temperatures are equal. Determine the spring constant of the spring.



(A) 200 N/m
(B) 20 N/m
(C) 2 N/cm
(D) 2 N/mm

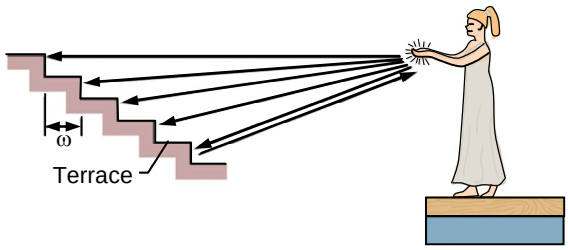
14. Heat flows from a reservoir at 373 K to a reservoir at 273 K through a copper rod as shown in the figure. The heat then leaves the 273 K reservoir and enters a Carnot engine, which uses part of this heat to do work and rejects the remainder to a third reservoir at 173 K. What fraction of the heat leaving the 373 K reservoir is rendered unavailable for doing work, as compared to the situation where a Carnot engine is connected directly between the 373 K and 173 K reservoirs?



(A) 3%
(B) 10%
(C) 17 %
(D) 35 %

Space for Rough work

15. A handclap on stage in an amphitheater sends out sound waves that scatter from terraces of width w = 0.75 m (see figure). The sound returns to the stage as a periodic series of pulses, one from each terrace; the parade of pulses sounds like a played note. Assuming that all the rays in Figure are horizontal, find the frequency at which the pulses return (that is, the frequency of the perceived note). Take the speed of sound to be 330m/s.



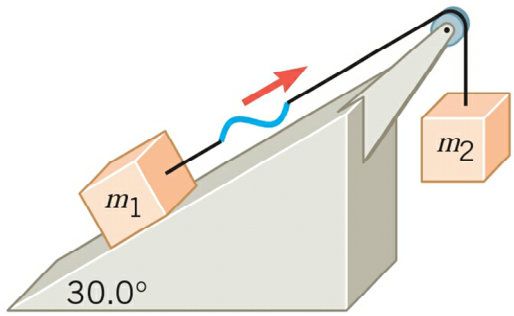
(A) 248Hz
(B) 220 Hz
(C) 456 Hz
(D) 440Hz

16. An RC- circuit, with R = 600kΩ and C= 10μF, is connected to a 5.0-V battery until the capacitor is fully charged. Then, the battery is suddenly replaced with a new 3.0-V battery of opposite polarity. At what time after this replacement will the energy stored in the capacitor be zero?

(given that $e \approx \frac{8}{3}$)

(A) 12 s
(B) 6 s
(C) 3 s
(D)1.5 s

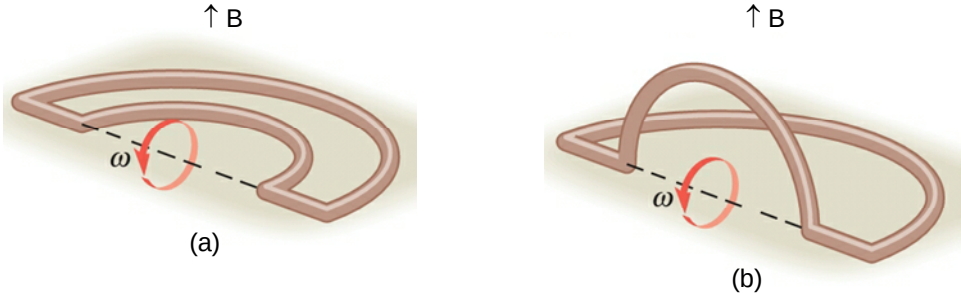
17. The drawing shows a frictionless incline and pulley. The two blocks are connected by a wire (mass per unit length, μ = 25 g/m) and remain stationary. A transverse wave on the wire has a speed of 60 m/s relative to it. Neglect the weight of the wire relative to the tension in the wire. If the mass m_2 be increased by 1%, the speed of the transverse wave will be



(A) 60.2 m/s
(B) 60.3 m/s
(C) 60.4 m/s
(D)60.6 m/s

Space for Rough work

18. The drawing shows a coil of copper wire that consists of two semicircles joined by straight sections of wire. In part (a) the coil is lying flat on a horizontal surface. The dashed line also lies in the plane of the horizontal surface. Starting from the orientation in part (a) the smaller semicircle rotates at an angular frequency ω about the dashed line, until its plane becomes perpendicular to the horizontal surface, as shown in part (b). A uniform magnetic field B, constant in time and is directed upward, perpendicular to the horizontal surface. The field completely fills the region occupied by the coil in either part of the drawing. The magnitude of the magnetic field is B= 2 T.



The resistance of the coil is 0.4 Ω, and the smaller semicircle has a radius of 20 cm while the bigger one has a radius of 40 cm. The angular frequency at which the small semicircle rotates is ω = 15 rad/s. Determine the average current I, if any, induced in the coil as the coil changes shape from that in part (a) of the drawing to that in part (b).

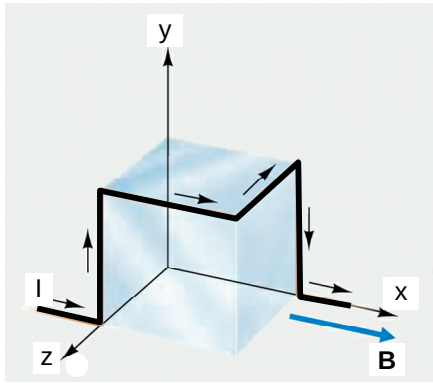
(A) 0 A. (B) 1A
(C)2A. (D) 3A

19. If a hole is punctured in a tire, the gas inside will gradually leak out of it. Let's assume the following: the area of the hole is A; the tire volume is V; and the time it takes for most of the air to leak out of the tire be t. This time can be expressed in terms of the ratio A/V, the temperature T, the Universal gas constant R, and the mass of the gas molecules inside the tire, m. Under these assumptions, we can use dimensional analysis to find an estimate for t. Assuming this estimate to be correct, if the mass of air within a tyre is increased by 70%, the absolute tyre temperature increased by 20%, while the area of the punctured hole is doubled (but still small) then the time in which a tyre will go flat will

(A)increase by 10% (B) increase by 25%
(C) decrease by 25%. (D) decrease by 40%

Space for Rough work

20. A long wire carries current of 25 A parallel to the positive x axis, except for three of the four segments that follow the edges of a cube of side 0.4 m, as shown in figure. The wire is in a uniform magnetic field of 2.0 T directed parallel to the positive x axis. What is the net magnetic force on the wire (in newton) ?



(A) $20\hat{j}$ (B) $20\hat{k}$

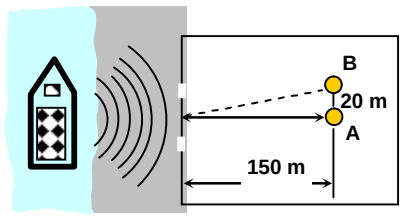
(C) $20(\hat{j}-\hat{k})$ (D) $-20\hat{j}$

21. A uniformly charged square plate having side L carries a uniform surface charge density σ. The plate lies in the y-z plane with its center at the origin. A point charge q lies on the x-axis. The flux of the electric field of q through the plate is ϕ_0; while the force on the point charge q due to the plate is F_0, along the x-axis. Then,

(A) $\sigma = \dfrac{F_0}{\phi_0 L}$ (B) $\sigma = \dfrac{F_0}{\phi_0}$

(C) $\sigma = \dfrac{F_0 L}{\phi_0}$ (D) $\sigma = \dfrac{\phi_0}{F_0}$

22. A riverside warehouse has two open doors as shown in Figure. Its walls are lined with sound-absorbing material. A boat on the river sounds its horn. To person A the sound is loud and clear. To person B the sound is barely audible. The principal wavelength of the sound waves is 2 m. Assuming person B is at the position of the first minimum, determine the distance between the doors, center to center. Distance AB = 20 m and the distance of A from the doors is 150 m.

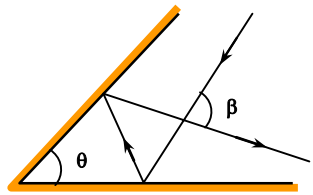


(A) 7.5 m (B) 15 m

(C) 5 m (D) 10 m

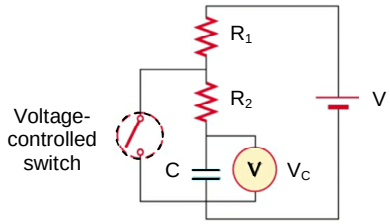
Space for Rough work

23. The reflecting surfaces of two intersecting flat mirrors are at an angle θ ($0° < \theta < 90°$), as shown in Figure. For a light ray that strikes the horizontal mirror, the emerging ray intersects the incident ray at an angle β. If one of the mirrors is rotated at $\frac{d\theta}{dt} = 1°/s$, then the angle β changes at the rate of

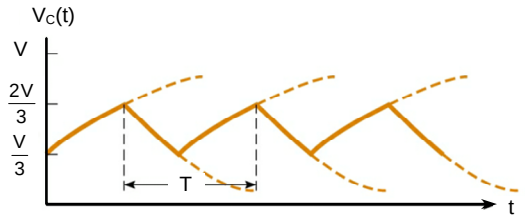


(A) 1°/s
(B) 2°/s
(C) –1°/s
(D) –2°/s

24. The switch in Figure (a) closes when Vc =2V/3 and opens when Vc = V/3. The voltmeter reads a voltage as plotted in figure (b). What is the period T of the waveform? (take R_1 = 6R and R_2 = 3R; and $\ell n2 = 0.7$ if required)



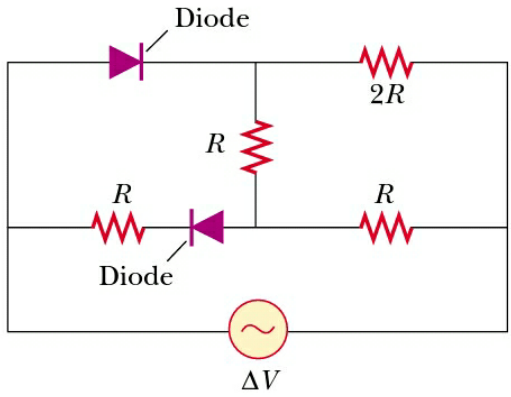
(a)



(b)

(A) 12 RC
(B) 8.4 RC
(C) 11 RC
(D) none of these

25. A diode is a device that allows current to be carried in only one direction (the direction indicated by the arrowhead in its circuit symbol). Find (approximately) in terms of ΔV and R the average power delivered to the diode circuit of the figure. Assume that the diodes are ideal and ΔV is the rms voltage applied to the circuit.



(A) $\frac{(\Delta V)^2}{R} \times 0.6$

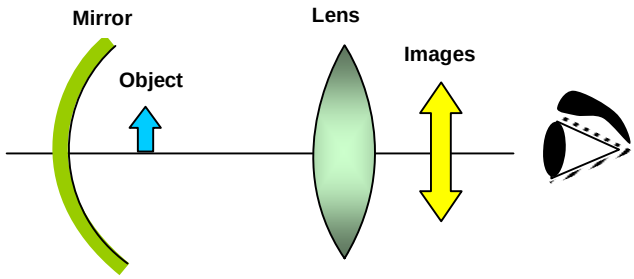
(B) $\frac{(\Delta V)^2}{R} \times 0.8$

(C) $\frac{(\Delta V)^2}{R} \times 1.1$

(D) $\frac{(\Delta V)^2}{R} \times 1.4$

Space for Rough work

26. An observer to the right of the mirror–lens combination shown in Figure sees two real images that are the same size and in the same location. One image is upright and the other is inverted. Both images are 1.50 times larger than the object. The lens has a focal length of 10.0 cm. The lens and mirror are separated by 40.0 cm. Determine the focal length of the mirror. Do not assume that the figure is drawn to scale.

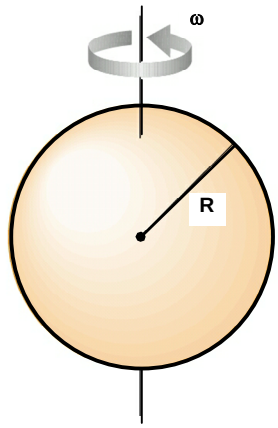


(A) 35/6 cm
(B) 35/3 cm
(C) 20 cm
(D) 10 cm

27. The angle of incidence of a light beam onto a reflecting surface is continuously varied. The reflected ray is found to be completely polarized when the angle of incidence is 60°. What is the index of refraction of the reflecting material?

(A) $\frac{\sqrt{3}}{2}$
(B) $\frac{2}{\sqrt{3}}$
(C) $\sqrt{3}$
(D) 2

28. A sphere of radius R has a uniform volume charge density. Determine the magnetic dipole moment of the sphere when it rotates as a rigid body with angular speed ω about an axis through its center. The total charge of the sphere is q.



(A) $\frac{2q}{3}R^2\omega$
(B) $\frac{q}{3}R^2\omega$
(C) $\frac{2q}{5}R^2\omega$
(D) $\frac{q}{5}R^2\omega$

Space for Rough work

29. A Rydberg atom is a single-electron atom with a large quantum number n. Rydberg states are close together in energy, so transitions between adjacent Rydberg states produce long-wavelength photons. Consider a transition from a state n + 1 to a state n in hydrogen. The wavelength of the emitted photon varies with quantum number n as

(A) $\frac{1}{n^2}$ (B) n

(C) $\frac{1}{n^3}$ (D) n^3

30. As part of his discovery of the neutron in 1932, James Chadwick determined the mass of the neutron (newly identified particle) by firing a beam of fast neutrons, all having the same speed, as two different targets and measuring the maximum recoil speeds of the target nuclei. The maximum speed arise when an elastic head-on collision occurs between a neutron and a stationary target nucleus. Represent the masses and final speeds of the two target nuclei as m_1, v_1, m_2 and v_2 and assume Newtonian mechanics applies. The neutron mass can be calculated from the equation:

$$m_n = \frac{m_1 v_1 - m_2 v_2}{v_2 - v_1}$$

Chadwick directed a beam of neutrons on paraffin, which contains hydrogen. The maximum speed of the protons ejected was found to be 3.3×10^7 m/s. A second experiment was performed using neutrons from the same source and nitrogen nuclei as the target. The maximum recoil speed of the nitrogen nuclei was found to be 4.7×10^6 m/s. The masses of a proton and a nitrogen nucleus were taken as 1 u and 14 u, respectively. What was Chadwick's value for the neutron mass?

(A) 1.16 u (B) 1.47 u

(C) 0.95 u (D) 1.80 u

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. A saturated solution of CaF_2 is found to contain 2×10^{-4} M of F^- ions. K_{sp} of CaF_2 is :
(A) 8×10^{-12} (B) 4×10^{-12}
(C) 2×10^{-12} (D) 10^{-12}

32. In the absence of external magnetic field, d-orbital is
(A) 3-fold degenerate (B) Non-degenerate
(C) 5-fold degenerate (D) 7-fold degenerate

33. Identify the correct statement:
(A) 0.1 M NaCl and 0.1 M glucose solution are isotonic.
(B) 0.1 M NaCl and 0.3 M glucose solution are isotonic.
(C) The boiling point of 0.1 m aqueous urea solution is less than 0.1 m aqueous KCl solution.
(D) The freezing point of 0.1 m glucose solution is less than 0.1 m KCl solution.

34. A reaction is carried out at 600 K. If the same reaction is carried out in the presence of catalyst at the same rate and same frequency factor, the temperature required is 500 K. What is the activation energy of the reaction, if the catalyst lowers the activation energy barrier by 20 kJ/mol?
(A) 100 (B) 120
(C) 80 (D) None of these

35. When pressure is applied to equilibrium system:
$\text{Ice} \rightleftharpoons \text{Water}$
Which of the following statement is correct?
(A) More amount of ice will form. (B) Melting point of ice decreases.
(C) Equilibrium will not disturb. (D) Both (A) and (B).

36. The enthalpy of neutralization of a strong acid by a strong base is – 57.32 kJ/mol. The enthalpy of formation of water is – 285.84 kJ/mol. The enthalpy of formation of aqueous hydroxyl ion is:
(A) +228.52 kJ/mol (B) -114.26 kJ/mol
(C) -228.52 kJ/mol (D) +114.2 kJ/mol

Space for Rough work

37. The ratio of coefficient of HNO_3, $Fe(NO_3)_2$ and NH_4NO_3 in the following redox reaction

$$Fe + HNO_3 \longrightarrow Fe(NO_3)_2 + NH_4NO_3 + H_2O$$

are respectively

(A) 10 : 1 : 4 (B) 4 : 10 : 1
(C) 4 : 1 : 10 (D) 10 : 4 : 1

38. An electron is revolving round the nucleus of He^+ ion with speed 2.2×10^6 m/s. The potential energy of the electron is (if atomic number of He = 2, 1 eV = 1.6×10^{-19} J. Mass of electron = 9×10^{-31} kg)

(A) - 13.61 eV (B) - 6.8 eV
(C) - 27.22 eV (D) zero

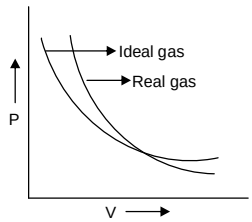
39. In a compound XY_2O_4, the oxide ions are arranged in CCP arrangement and cations X are present in octahedral. Cations Y are equally distributed between octahedral and tetrahedral voids. The fraction of the octahedral voids occupied is

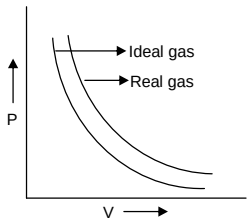
(A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{1}{6}$ (D) $\frac{1}{8}$

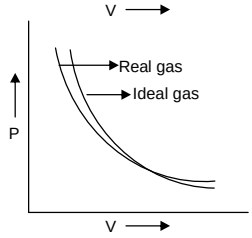
40. The conductivity of saturated solution of $Ba_3(PO_4)_2$ is $1.2\times10^{-5}\Omega^{-1}cm^{-1}$. The limiting equivalent conductivities of $BaCl_2$, K_3PO_4 and KCl are 160, 140 and 100 $\Omega^{-1}cm^2eq^{-1}$, respectively. The solubility product of $Ba_3(PO_4)_2$ is

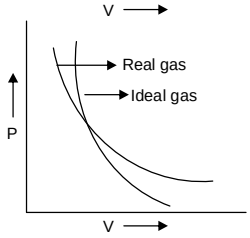
(A) 10^{-5} (B) 1.08×10^{-23}
(C) 1.08×10^{-25} (D) 1.08×10^{-27}

41. Which of the following graph is correct for real gases other than hydrogen and helium at 0°C?

(A)


(B)


(C)


(D)


Space for Rough work

42. $[Co(NH_3)_4(NO_2)_2]Cl$ exhibits
(A) linkage isomerism, geometrical isomerism and optical isomerism.
(B) linkage isomerism, ionisation isomerism and optical isomerism.
(C) linkage isomerism, ionisation isomerism and geometrical isomerism.
(D) ionisation isomerism, geometrical isomerism and optical isomerism.

43. In which of the following species $p\pi - d\pi$ bond is present but $p\pi - p\pi$ bond is absent?
(A) SO_3 (B) SO_2
(C) CS_2 (D) SO_2Cl_2

44. Which of the following statement is incorrect about the structure of B_2H_6 is
(A) It has four $2C - 2e^-$ bond and two $3C - 2e^-$ bond.
(B) The hybridization of each boron atom is sp^3.
(C) All hydrogens in B_2H_6 lie in the same plane.
(D) Terminal bonds are shorter than bridged bonds.

45. Which of the following can't be separated by H_2S in dil. HCl?
(A) Bi^{3+}; Mn^{2+} (B) Pb^{2+}; Hg^{2+}
(C) Zn^{2+}; Cu^{2+} (D) Ni^{2+}; Cu^{2+}

46. The correct order of increasing oxidizing power in the following series is:
(A) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ (B) $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$
(C) $Cr_2O_7^{2-} < MnO_4^- < VO_2^+$ (D) $MnO_4^- < Cr_2O_7^{2-} < VO_2^+$

47. Aqueous solution of $Na_2S_2O_3$ on reaction with Cl_2 gives
(A) $Na_2S_4O_6$ (B) $NaHSO_4$
(C) NaCl (D) NaOH

48. Which one of the following is an incorrect statement?
(A) Ozone is diamagnetic in nature. (B) Ozone oxidizes PbS to $PbSO_4$.
(C) Ozone oxidizes acidic solution of KI. (D) Ozone is used an oxidizing agent in the manufacture of $KMnO_4$.

49. Which of the following statement is correct?
(A) LiOH > NaOH > KOH > RbOH (solubility order)
(B) $Be(OH)_2 > Mg(OH)_2 > Ca(OH)_2 > Sr(OH)_2$ (basic strength order)
(C) Li < K < Na < Rb < Cs (density order)
(D) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ (ionic mobility in aqueous medium)

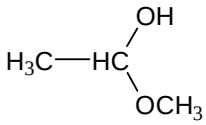
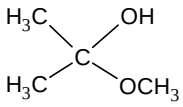
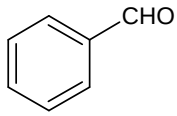
50. Coagulation is not done by
(A) Persistant dialysis (B) Boiling
(C) Electrophoresis (D) Peptisation

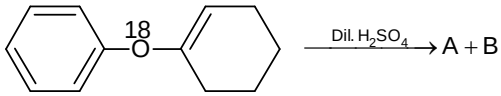
Space for Rough work

51. $F_2C = CF_2$ is a monomer of
(A) Nylon (B) Teflon
(C) Glyptal (D) Buna-S

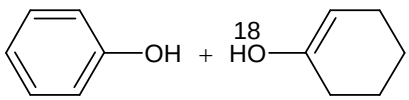
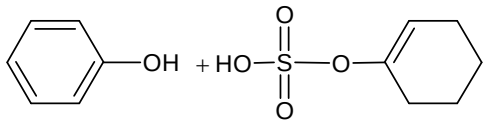
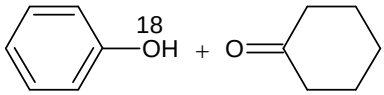
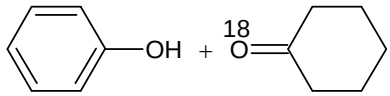
52. Which of the following test is used for identifying carbohydrates?
(A) Millon's test (B) Molisch's test
(C) Biuret test (D) Nihydrin test

53. Which of the following compound does not give Tollen' test?

(A) 
(B) H – COOH
(C) 
(D) 

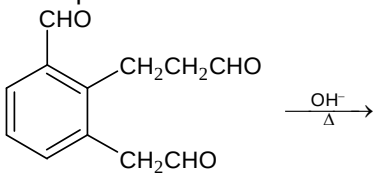
54. 

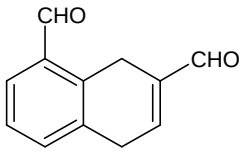
Identify the major products of the above reaction?

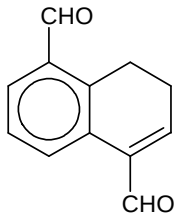
(A) 
(B) 
(C) 
(D) 

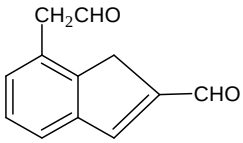
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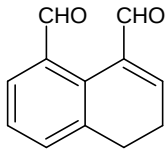
55. Which product can't be obtained in following reaction?



(A) 

(B) 

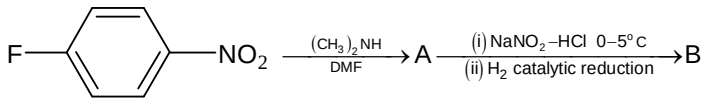
(C) 

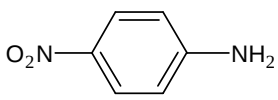
(D) 

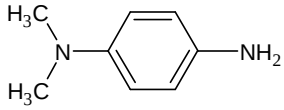
56. Which of the following orders is true regarding the acidic nature of phenols?

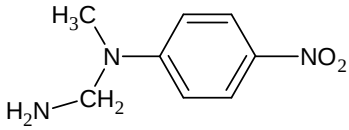
(A) Phenol > o-cresol < o-nitrophenol (B) Phenol < o-cresol < o-nitrophenol

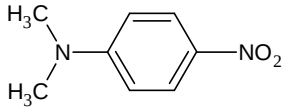
(C) Phenol > o-cresol > o-nitrophenol (D) phenol < o-cresol > o-nitrophenol

57. 

(A) 

(B) 

(C) 

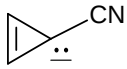
(D) 

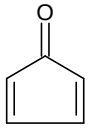
Space for Rough work

58. When benzamide is treated with $POCl_3$ the product is
(A) Aniline (B) Chlorobenzene
(C) Benzylamine (D) Benzonitrile

59. Which of the following species is aromatic?

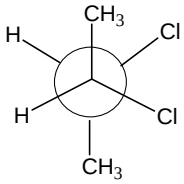
(A) 

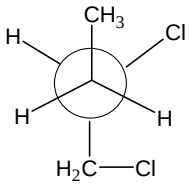
(B) 

(C) 

(D) 

60. The following pair is related as





(A) Same compound (B) Diastereomers
(C) Enantiomers (D) Structural isomers

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Sum of series of $\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+.....\infty$ is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

62. From any point R two normals which are right angled to one another are drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (a > b). If the feet of the normals are P and Q then the locus of the circumcentre of the ΔPQR is

(A) $\frac{x^2+y^2}{a^2-b^2}=\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2$ (B) $\frac{x^2-y^2}{a^2-b^2}=\left(\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)^2$

(C) $\frac{x^2+y^2}{a^2-b^2}=\left(\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)^2$ (D) $\frac{x^2+y^2}{a^2+b^2}=\left(\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)^2$

63. If point P lies on the plane of ΔABC such that $\angle$BAC = $\angle$BPC and H is orthocentre of ΔABC. D is midpoint of BC and E is the midpoint of PH, then

(A) DE is perpendicular to AP (B) DE is parallel to AH

(C) DE is perpendicular to PB (D) none of these

Space for rough work

64. Let f: [0, 1] $\to$ R be a continuous function then which of the following is maximum value of $\int_0^1 x^2 f(x)\,dx - \int_0^1 x\left(f(x)\right)^2 dx$ for all such functions

(A) $\frac{1}{12}$ (B) $\frac{1}{16}$

(C) $\frac{1}{8}$ (D) $\frac{1}{20}$

65. If f(x) is a polynomial function with positive integral coefficients (degree of f(x) $\geq$ 1) and for x $\geq$ 1, $\frac{f\left(f[x]+1\right)}{f\left([x]\right)}$ is an integer then (where [.] denotes the greatest integer function)

(A) x $\in$ [1, 2) (B) x $\in$ [2, 3)

(C) x $\in$ [3, 4) (D) none of these

66. A five digits number has to be formed by using the digits 1, 2, 3, 4 and 5 without repetition such that the even digits occupy odd places. Find the sum of all such possible numbers

(A) 1199988 (B) 1166688

(C) 1266688 (D) none of these

67. Suppose two complex numbers z = a + ib, ω = c + id satisfy the equation $\frac{z+\omega}{z} = \frac{\omega}{z+\omega}$ then

(A) both a and c are zero (B) both b and d are zero

(C) both b and d must be non zero (D) at least one of b and d is non zero

68. Let OP, OQ, OR are three edges of a regular tetrahedron of edge length a. If $\vec{p}, \vec{q}$ and $\vec{r}$ are the position vectors of the points P, Q and R & O is the origin then $\left|\vec{p}\times\vec{q}+\vec{q}\times\vec{r}+\vec{r}\times\vec{p}\right|$ is equal to

(A) $\frac{\sqrt{3}a^2}{2}$ (B) $\frac{a^2}{2}$

(C) $\frac{3\sqrt{3}a^2}{2}$ (D) $\frac{7\sqrt{3}a^2}{3}$

69. Find the number of terms in the expansion of $(x + y + z)^6$ which must contain x

(A) 28 (B) 24

(C) 22 (D) 21

Space for rough work

70. If $\vec{a}, \vec{b}, \vec{c}$ are three non coplanar unit vectors, the angle between them pair wise are $\frac{\pi}{6}, \frac{\pi}{4}$ and $\frac{\pi}{3}$ then the $\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$ is

(A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(C) $\frac{\sqrt{2+\sqrt{6}}}{2}$ (D) $\frac{\sqrt{\sqrt{6}-2}}{2}$

71. If $\omega \neq 1$ is cube root of unity and $x + y + z \neq 0$, then $\begin{vmatrix} \frac{x}{1+\omega} & \frac{y}{\omega+\omega^2} & \frac{z}{\omega^2+1} \\ \frac{y}{\omega+\omega^2} & \frac{z}{\omega^2+1} & \frac{x}{1+\omega} \\ \frac{z}{\omega^2+1} & \frac{x}{1+\omega} & \frac{y}{\omega+\omega^2} \end{vmatrix}$ is equal to zero if

(A) $x^2 + y^2 + z^2 = 0$ (B) $x + yz + z\omega^2 = 0$ or $x = y = z$

(C) $x \neq y \neq z \neq 0$ (D) $x = 2y = 3z$

72. If $a_1, a_2, a_3, \ldots\ldots a_{2n+1}$ are in A.P., then $\frac{a_{2n+1}-a_1}{a_{2n+1}+a_1}+\frac{a_{2n}-a_2}{a_{2n}+a_2}+\ldots..+\frac{a_{n+2}-a_n}{a_{n+2}+a_n}$ is equal to

(A) $\frac{n(n+1)}{2}\frac{(a_2-a_1)}{a_{n+1}}$ (B) $\frac{n(n+1)}{2}$

(C) $\frac{(n-1)n(a_2-a_1)}{a_{n+1}}$ (D) $\frac{n(n-1)(n+1)(a_2-a_1)}{a_{n+1}}$

73. ABC is right triangle in which $\angle B = 90^o$ and BC = a. If n points $L_1, L_2, L_3, \ldots\ldots, L_n$ on AB is divided in (n + 1) equal parts and $L_1M_1, L_2M_2, L_3M_3, \ldots\ldots, L_nM_n$ are line segments parallel to BC and $M_1, M_2, M_3, \ldots\ldots, M_n$ are on AC, then the sum of the lengths $L_1M_1, L_2M_2, \ldots\ldots, L_nM_n$ is

(A) $\frac{a(n+1)}{2}$ (B) $\frac{a(n-1)}{2}$

(C) $\frac{an}{2}$ (D) none of these

Space for rough work

74. Find the sum of the series $\frac{{}^nC_0}{2.3}-\frac{{}^nC_1}{3.4}+\frac{{}^nC_2}{4.5}-\frac{{}^nC_3}{5.6}+.....+(-1)^n\frac{{}^nC_n}{(n+2)(n+3)}$

(A) $\frac{1}{(n+3)(n+2)}$ (B) $\frac{1}{(n+1)(n+2)(n+3)}$

(C) $\frac{2}{(n+1)(n+2)}$ (D) $\frac{2}{(n+1)(n+2)(n+3)}$

75. If x + 2y + 3z = n and x, y, z are positive integers then the number of ordered trip lets (x, y, z) satisfying the equation are

(A) ${}^nC_{n-6}$ (B) ${}^nC_{n-3}$

(C) ${}^{n+2}C_3$ (D) none of these

76. px + qy = 40 is a chord of minimum length of circle $(x-10)^2+(y-20)^2=729$. If the chord passes through (5, 15), then $p^{2013}+q^{2013}$ is equal to

(A) 0 (B) 2

(C) 2^{2013} (D) 2^{2014}

77. The area enclosed by the curve max{|x – 1|, |y|} = k is 100, then k is equal to

(A) 5 (B) 8

(C) 10 (D) none of these

78. The value of $\lim\limits_{x\to\infty} x^2\ell n\left(x\cot^{-1}x\right)$ is

(A) $-\frac{1}{3}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{-2}{3}$

79. $\int\frac{\cos x+\sqrt{3}}{1+4\sin\left(x+\frac{\pi}{3}\right)+4\sin^2\left(x+\frac{\pi}{3}\right)}dx$ is

(A) $\frac{\cos x}{1+2\sin\left(x+\frac{\pi}{3}\right)}+c$ (B) $\frac{\sec x}{1+2\sin\left(x+\frac{\pi}{3}\right)}+c$

(C) $\frac{\sin x}{1+2\sin\left(x+\frac{\pi}{3}\right)}+c$ (D) $\frac{1}{2}\tan^{-1}\left(1+2\sin\left(x+\frac{\pi}{3}\right)\right)+c$

Space for rough work

80. The minimum value of the function $f(x) = \frac{1}{2\{-x\}} - \{x\}$ occurs at x equal to

(A) $n + \frac{1}{\sqrt{2}}, n \in I$ (B) $n - \frac{1}{\sqrt{2}}, n \in I$

(C) $n + \frac{1}{2\sqrt{2}}, n \in I$ (D) $n + \frac{1}{2}, n \in I$

81. $\int \frac{x^3 - x^2 + x - 1}{x^5 + 1} dx$ is

(A) $\frac{1}{5}\ell n\left(\frac{1+x^5}{(1+x^3)^5}\right) + c$ (B) $\frac{1}{5}\ell n\left(\frac{1+x^5}{1+x^3}\right) + c$

(C) $\frac{1}{5}\ell n\left(\frac{1+x^5}{(1+x)^5}\right) + c$ (D) $\frac{1}{5}\ell n\left(\frac{1+x^5}{1+x}\right) + c$

82. If A is skew symmetric matrix and n is odd positive integer then A^n is

(A) a symmetric matrix (B) a skew symmetric matrix

(C) a diagonal matrix (D) none of these

83. A random variable x has probability distribution

x	1	2	3	4	5	6	7	8
P(x)	0.13	0.22	0.12	0.21	0.13	0.08	0.06	0.05

For the event E = {x is an odd number}, F = {x is divisible by 3} and G = {x is less than 7} the probability $P(E \cup (F \cap G))$ is equal to

(A) 0.87 (B) 0.77

(C) 0.52 (D) none of these

84. A plane containing the line $\frac{x-3}{2} = \frac{y-1}{4} = \frac{z-2}{5}$ and it is passing through the point (4, 3, 7). The equation of the plane is

(A) 4x + 8y + 8z = 4 (B) 4x – 8y – 8z = 4

(C) 4x – 2y – 10 = 0 (D) 4x – 8y + 8z = 4

Space for rough work

85. If $f(x) = \sin^4 x + \cos^4 x - \frac{1}{2}\sin 2x$ then the range of f(x) is

(A) $\left[0, \frac{3}{2}\right]$ (B) $\left[-\frac{1}{2}, \frac{7}{2}\right]$

(C) $\left[0, \frac{9}{8}\right]$ (D) $\left[\frac{3}{4}, \frac{7}{8}\right]$

86. The exhaustive interval of real values of x such that $\sqrt{12-4x} > 1+\sqrt{4x+4}$ is

(A) $\left[1-\frac{\sqrt{31}}{8}, 1+\frac{\sqrt{31}}{8}\right)$ (B) $\left[-1, 1+\frac{\sqrt{31}}{8}\right)$

(C) [–1, 3] (D) $\left[-1, 1-\frac{\sqrt{31}}{8}\right)$

87. Let 1, α, α^2, α^3, ..., α^{n-1} are nth roots of unity and for a natural number P, $A_P = \left\{x : x = \left(\alpha^P\right)^k\right\}$, $k \in I$, then

(A) A_P has n elements if P is a divisor of n
(B) number of elements in A_P is one less than the highest proper divisor of n (if P is divisor of n)
(C) number of elements in A_P is always a divisor of n
(D) none of these

88. Triangle ABC is formed by the lines 7x – y + 3 = 0, x + y – 3 = 0 and x – 3y – 31 = 0. Three new lines are drawn parallel to BC, CA, AB passing through A, B and C intersecting each other at P, Q and R respectively the circum centre of ΔPQR lies on

(A) 6x + 2y = 5 (B) 3x + y = 3
(C) 5x + y = 0 (D) none of these

89. If atleast one root of quadratic equation with integral coefficients $ax^2 + bx + c = 0$ (where a, b and c are sides of a triangle) is same as one of the roots of the quadratic equation $x^2 + kx + 2 = 0$ (k > 12), then $\tan\left(\frac{A+C}{2}\right)\cot\left(\frac{C-A}{2}\right)$ is equal to

(A) 1 (B) $\sqrt{2}$

(C) $\sqrt{3}$ (D) 3

90. If $f''(x) + f'(x) + f^2(x) = x^2$ be differential equation of a curve and let P be the point of maxima then the number of tangents which can be drawn from P to $x^2 - y^2 = a^2$ is/are

(A) 1 (B) 2
(C) 3 (D) none of these

Space for rough work